PDPChain: A Consortium Blockchain-Based Privacy Protection Scheme for Personal Data

Wei Liang, Yang Yang, Ce Yang, Yonghua Hu, Songyou Xie, Kuan-Ching Li, *Senior Member, IEEE*, and Jiannong Cao, *Fellow, IEEE*

***Abstract*—With the advances and innovations in digital technologies, blockchain has empowered advancements in communications and networking, promising to build trust and establish secure decentralized communications networks. Unfortunately, current personal data privacy protection schemes still suffer from explicit storage, lack of data ownership and implementation of fine-grained access control by users, and lack of transparency and auditability of data. In this article, we propose a personal data privacy protection scheme based on consortium blockchain that stores original data encrypted with an improved Paillier homomorphic encryption mechanism, namely PDPChain, where users realize fine-grained access control based on ciphertext policy attribute-based encryption (CP-ABE) on blockchain. In this scheme, consortium blockchain combines distributed private clusters to store the encrypted data, improving data transmission efficiency, and guaranteeing user privacy and security through off-chain storage and on-chain transmission synergy. In addition, it is more lightweight encryption and demarcation, ultimately protecting personal data privacy and providing a secure and trusted way to obtain information for data mining. For the performance testing, data in the form of files are used as an example, and the scheme is designed and simulated on Hyperledger Fabric and InterPlanetary File System. Experimental results show that the improved Paillier encryption mechanism reduces the overall encryption and decryption elapsed time by 25% and encryption elapsed time by 48%. Furthermore, the proposed CP-ABE access control method is adaptive to storing and sharing a massive amount of data. With the increase in the number of access control policies, the overall time-consuming of the scheme does not increase, and the time-consuming of decryption can also be stabilized at about 2 s.**

***Index Terms*—Consortium blockchain, ciphertext policy attribute-based encryption (CP-ABE), homomorphic encryption, InterPlanetary File System (IPFS), privacy protection, secure sharing.**

I. Introduction

With the advances of artificial intelligence (AI) and explosion in the huge amount of personal Internet data [1] and Internet-of-Things (IoT) industrial data [2], [3], the privacy protection of data has become a current research hotspot. Privacy protection and safe sharing are the core of data governance driven [4]. Traditional third-party cloud servers are used to store users' data, which is used as a dataset for training machine learning models, analyzing user preferences, or evaluating user credit [5]. Therefore, it is insecure for users to store their private data in third-party cloud servers because of the risk of data leaks. Thus, in the AI era [6], security data storage has attracted considerable critical attention [7].

As abovementioned, insecure centralized processing and explicit storage in third-party services in traditional cloud-based storage schemes is a pressing issue in academia [8]. As of today, personal data are mainly stored in the cloud [9], [10]; however, there are certain risks associated with the use of third-party cloud storage services, e.g., malicious tampering, data loss, and hacker attacks [11]. In addition, users lack ownership of the data, as they may belong to the third-party service provider once signed up, and users lack the right to know and control the private data [1], [4]. Besides, data lacks transparency and audibility as third-party service companies are semitrusted and manage data in the cloud servers [12], [13]. With the application of technologies such as deep reinforcement learning (DRL), unauthorized third parties may utilize users' private data to generate large amounts of training data, posing a threat to data through powerful data mining and analysis. Furthermore, users cannot implement fine-grained access control to data [8], because users do not clearly understand what data are accessed, mined, and analyzed.

In response to the abovementioned problems, one of the most significant challenges is how to achieve a secure and reliable method for personal data protection and sharing. Among the proposed approaches, blockchain has emerged as one of the most effective solutions to the challenge. As a decentralized

Manuscript received 10 April 2022; revised 25 May 2022; accepted 10 July 2022. Date of publication 5 August 2022; date of current version 5 June 2023. This work was supported in part by the National Key Research and Development Program of China under Grant 2021YFA1000600, in part by the National Natural Science Foundation of China under Grant 62072170, in part by the Science and Technology Project of Department of Communications of Hunan Provincial under Grant 202101, in part by the Key Research and Development Program of Hunan Province under Grant 2022GK2015, in part by the Hunan Provincial Natural Science Foundation of China under Grant 2021JJ30141, and in part by the Hunan Provincial Department of Education under Grant 20B242. Associate Editor: W. Chu. *(Corresponding author: Kuan-Ching Li.)*

Wei Liang is with the College of Computer Science and Electronic Engineering, Hunan University, Changsha 410082, China, with the School of Computer Science and Engineering, Hunan University of Science and Technology, Xiangtan 411201, China, and also with the Hunan Key Laboratory for Service Computing and Novel Software Technology, Xiangtan 411201, China (e-mail: wliang@hnust.edu.cn).

Yang Yang and Songyou Xie are with the College of Computer Science and Electronic Engineering, Hunan University, Changsha 410082, China (e-mail: yangyang0607@hnu.edu.cn; syxie@hnu.edu.cn).

Ce Yang, Yonghua Hu, and Kuan-Ching Li are with the School of Computer Science and Engineering, Hunan University of Science and Technology, Xiangtan 411201, China, and also with the Hunan Key Laboratory for Service Computing and Novel Software Technology, Xiangtan 411201, China (e-mail: yangce@hnust.edu.cn; huyh@hnust.edu.cn; rob5li@outlook.com).

Jiannong Cao is with the Department of Computing, Hong Kong Polytechnic University, Hong Kong (e-mail: csjcao@comp.polyu.edu.hk).

Color versions of one or more figures in this article are available at https://doi.org/10.1109/TR.2022.3190932.

Digital Object Identifier 10.1109/TR.2022.3190932

0018-9529 © 2022 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: SOUTHWEST JIAOTONG UNIVERSITY. Downloaded on March 01,2026 at 07:20:00 UTC from IEEE Xplore. Restrictions apply.

distributed ledger, blockchain can realize traceable, tamper-proof, and sharable transactions [14], providing novel solutions to problems of single point of failure (SPOF) and data tampering that exist in the explicit storage of personal data. Blockchain can be classified into permissionless blockchain, and permissioned blockchain, based on whether newly added nodes require authentication [15]. The popular Ethereum represents permissionless blockchain as public blockchain [16]. With public blockchain combined with cloud storage solutions [8], [12], users can freely join or exit the blockchain without previous authorization; however, this leads to inefficient execution and a relatively high cost of access to the blockchain. Permissioned blockchain can be classified into consortium blockchain, and private blockchain, based on the number of participants [14]. Private blockchain is mainly for single users, so it is not suitable for personal data sharing scenarios. In contrast, consortium blockchain provides channel-based data isolation protection by integrating authentication mechanism, and multi-channel technology [17], and only authorized nodes that pass the authentication mechanism can join the blockchain network [14]. In the consortium blockchain, all users jointly maintain the blockchain network, balancing efficiency and fairness, reducing energy consumption, and increasing efficiency, allowing for access control to data and better privacy protection.

There is a huge potential for solving privacy protection issues with the application of blockchain technology [12], [18]–[23]. In scheme [12], Zyskind et al. designed a distributed personal data management system based on blockchain to give users ownership and control over their data and to propose several possible extensions for the future development of the blockchain. Privacy protection of medical data has always been a hot research topic, among which MedRec is a typical case of electronic health record (EHR) management and medical data, proposing a distributed and decentralized patient health data management system [18]. Li et al. combined homomorphic encryption and attribute-based cryptography based on blockchain for medical data storage and reliable sharing [20]. However, the scheme lacks ease of use and also data cannot be shared and transmitted between different institutions. Tan et al. proposed a blockchain-based approach for privacy protection and secure data sharing of COVID-19 medical records [23]. Regrettably, these schemes only provide preliminary methodological exploration or model concepts without detailed technical analysis and implementation, so the problem-solving scenarios are relatively homogeneous. They do not realize the problems faced by all scenarios of personal privacy data. At the same time, although the consortium blockchain can provide data isolation protection based on the channel level, this data protection mechanism still synchronizes plaintext data in the channel, so there is a certain degree of data leakage risk. Although plaintext encryption of data is achieved through improved cryptography with heap into, only the user with the key can decrypt it, and fine-grained access control cannot be achieved [24]. In general, there is a lack of a comprehensive framework and model to address the issues faced by personal data privacy protection.

We positively answer these problems by presenting PDPChain, a fine-grained data access control method based on ciphertext policy attribute-based encryption (CP-ABE) [25], which achieves on-chain and off-chain cooperative storage through consortium blockchain combined with distributed private clusters, and data security protection combined with Paillier homomorphic encryption mechanism [26]. This article makes the following key contributions.

1) To propose a personal data privacy protection method based on a consortium blockchain, combined distributed applications to realize users' visual storage and tracking of privacy data.
2) To encrypt data stored based on distributed private clusters. Data owners can efficiently and securely store data in a decentralized privacy cluster, forming an on-chain and off-chain cooperative storage, saving the memory burden of on-chain nodes, and realizing the protection, verification, supervision, and sharing of data information.
3) To use CP-ABE access control method and certificate authority (CA) technologies and collaboratively solve the user's access control to the data. The data owner can set access control policies by the identity attributes of other users. Only users who conformed to the policies can access the data.
4) To employ the Paillier homomorphic encryption mechanism and accomplish data transmission on blockchain in ciphertext rather than plaintext. Simultaneously, data can be securely applied to scenarios such as cloud computing, machine learning, and federated learning to achieve usable invisibility of the data.

The rest of this article is organized as follows: Related work is presented in Section II while the preliminaries of the proposed PDPChain are in Section III. Next, details and approaches of the proposed PDPChain, including the overall design, system components, system flow, and smart contract design, are depicted in Section IV. To further demonstrate the proposed PDPChain, experimental results and analysis are shown in Section V, and finally, Section VI concludes this article.

II. Related Work

With the rapid development of blockchain technology, much scientific research and institutional personnel has explored promising practices of privacy data protection. Most of the current research work is mainly based on blockchain technology to address issues related to explicit storage of data, data ownership, data transparency and audibility, and fine-grained access control of data.

1) *Explicit storage of data:* Due to issues such as lack of transparency and trust in centralized or decentralized cloud storage solutions, a great deal has been done to solve the problem of storing data through emerging technologies of blockchain. The consortium blockchain enhances data privacy protection through a multichannel data isolation mechanism [14], but the data within the channel are still visible to the nodes in the same channel. To protect the privacy of data on the blockchain, Ma et al. [27] proposed a NIZK scheme based on account-model blockchain that hides the account balance using homomorphic encryption.

Authorized licensed use limited to: SOUTHWEST JIAOTONG UNIVERSITY. Downloaded on March 01,2026 at 07:20:00 UTC from IEEE Xplore. Restrictions apply.

Bünz et al. proposed Zether, a de-trust mechanism for privacy-preserving payments in smart contract platforms. Zether ensures data privacy by encrypting account balances, transaction addresses, and information [28].

2) *Data ownership:* The moment a user uploads private data to the cloud server, he may lose ownership of the data, and therefore, the security and privacy of the data cannot be guaranteed. Several studies have examined how to maintain data ownership for users. Liu et al. proposed a secure data sharing mechanism combining cloud, CP-ABE, and consortium blockchain [22], where the ownership and management of data are controlled by specific personnel such as administrators, and general users cannot manage and control their data. Gao et al. combined blockchain, CP-ABE, and InterPlanetary File System (IPFS) technologies to design a user-centric secure data sharing scheme for BSSPD [4]. Nonetheless, it can only guarantee the security of the data in sharing, not in the storage process. Piao et al. proposed a service-on-chain (SOC) approach to ensure privacy protection, trusted data content, and controllable data ownership for government data-sharing [29]. Nevertheless, the main problem solved is only the ownership attribution in government data sharing.
3) *Fine-grained access control of data:* Access control technologies are used to authorize and manage user permissions. The blockchain-based access control methods realize identity authentication and authority access control mechanisms, which can improve the scalability and security of the system due to the public verifiability and immutability of blockchain. Role-based and attribute-based access control approaches are widely used access control solutions due to secure identity division, and cryptography, such as attribute-based access control and role-based access control. Role-based access control methods assign appropriate permissions based on roles and are now primarily used to address the management and protection of medical data [18]. Several research works solved the security problem of cloud storage by combining blockchain technology and attribute-based encryption to achieve fine-grained access control of data [30]. Qiu et al. proposed a data access control scheme based on blockchain and CP-ABE encryption [31]. Liu et al. realized a data management framework based on blockchain and searchable attribute-based encryption (BS-SABE), which can store and process massive IoT data [22]. However, users need to create partial tokens through consortium blockchain, and the system cannot store large files. Poongodi et al. combined encryption and hashing techniques to provide a reliable keyword search of encrypted files in decentralized storage [32]. To improve privacy, Guan et al. proposed a blockchain-based privacy-preserving distributed energy trading scheme to protect user privacy [33]. Alternatively, although the above schemes achieve secure data sharing, they cannot guarantee well for the trusted storage of data [18].
4) *Data transparency and auditability:* To enable each user to understand how their data are handled and managed fully, most current approaches are based on blockchain and homomorphic encryption to design solutions. Zhang et al. achieved blockchain-based data auditing scheme for multicloud storage to protect data integrity and accurately arbitrate service disputes [13]. Although transaction records are all stored on blockchain, it may bring a burden to the storage overhead on the blockchain. Li et al. designed the EHRChain electronic medical record management system based on homomorphic encryption technology [20]. During the medical claim process, doctors and insurance companies use ciphertext to check patient information, ensuring the privacy and security of patient data. However, most current solutions are aimed at a single application scenario and cannot provide data transparency and auditability for all users' private data. The original data are stored and shared on the blockchain in plaintext in specific scenarios.

In short, there are still defects and shortcomings when it comes to the privacy protection of personal data in blockchain, which cannot realize the purpose of credible storage and secure sharing of private data. In contrast, our approach can achieve the objective by combining consortium blockchain, CP-ABE, Paillier, and distributed private cluster.

III. Preliminaries

In this section, we describe the blockchain technology, the improved Paillier homomorphic encryption mechanism, and the improved CP-ABE access control method in the PDPChain scheme.

A. Blockchain as the Backbone for Privacy-Preserving Information Sharing

The consortium blockchain is a polycentric and partially decentralized blockchain where only authorized nodes that have passed the authentication mechanism can join the blockchain network [14]. All users jointly maintain the blockchain network, which allows easy access control of data and better privacy protection. The consortium blockchain has the advantage of more secure storage, better efficiency, and greater throughput compared to public blockchain.

The consortium blockchain is a blockchain technology suitable for multiparty collaboration and consensus, and the multichannel technology of the consortium blockchain can establish different organizations for collaborative management while configuring a CA to manage the authorized identity of users. CA in the scheme is a third-party trusted module, a CA trusted by all nodes on blockchain, mainly used to generate the identity proof of the nodes on blockchain. CA provides three functions, one is to register the identity of new users, and the registration result will be used as the credentials of the user's certificate application. The second is to issue certificates, i.e., to generate certificates for the legal registration credentials. The third is to update or revoke certificates, and this function will update the corresponding user's certificate information. Also, in the PDPChain, the RAFT consensus protocol [34] is used because it can work without affecting the system in case of partial node

Authorized licensed use limited to: SOUTHWEST JIAOTONG UNIVERSITY. Downloaded on March 01,2026 at 07:20:00 UTC from IEEE Xplore. Restrictions apply.

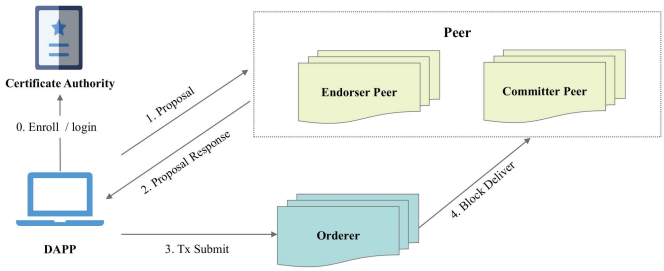


Fig. 1. Blockchain transaction process.

crashes. In the scenario of personal data privacy protection and secure sharing, there is a strong consistency requirement for consensus and a higher requirement for consensus efficiency. Since RAFT is based on the consistency algorithm of message passing, it is more efficient in the practical scenario of the distributed system. So, this article uses RAFT for consistent consensus in a blockchain network.

As Fig. 1 shows the whole transaction process of blockchain from issuing a transaction to final accounting. The client distributed applications (DAPP) initiates a transaction through the SDK, registers with the CA, and obtains an ID certificate. After receiving the information, the DAPP packages the data into transactions and sends them to the Orderer, who sorts the received transactions by consensus and generates a new block to the committer peer, who receives the block, checks each transaction in the block, synchronizes the block, and updates the ledger.

B. Based on Paillier Homomorphic Encryption Mechanism

Homomorphic encryption can perform any calculation on encrypted data without decrypting the ciphertext data. Applying homomorphic encryption to cloud services may solve the problem of confidential storage and confidential computation of data in cloud services.

As per (1), homomorphic encryption is addable. Moreover, in additive homomorphic encryption, x and y are not leaked

$$\text{Enc}(x+y) = \text{Enc}(x) + \text{Enc}(y). \tag{1}$$

To improve the design of the Paillier homomorphic encryption scheme [35], Paillier referenced the probabilistic encryption scheme based on the decision and number residual problem [26] and built over the improved Paillier public key probabilistic encryption system. The scheme consists of four algorithms, as follows.

PKenGen: The public key $n = pq$, where $n = pq$ is the RSA modulus, input two large prime numbers p and q and then calculate $n = pq$. The secret key $b = \lambda^{-1} \text{mod} n$, where $\lambda = \text{lcm}(p-1, q-1)$. $\text{lcm}(p-1, q-1)$ means to find the least common multiple of $p-1$ and $q-1$.

PEncrypt: In the encryption process, r is chosen randomly before encryption. The encryption user can randomly select more than one before encryption and then perform the preoperation $r^n \text{mod} n^2$. In the encryption process, the user only needs to randomly select some $r^n \text{mod} n^2$ to encrypt the plaintext M. From (2), it can be seen that encryption only requires two simple multiplication operations on Z_{n^2}, which further improves the efficiency of encryption

$$\text{CT} = (1+Mn)r^n \bmod n^2 \tag{2}$$

where $M \in Z_n$ is the data to be encrypted.

PDecrypt: Using the private key to decrypt, the plaintext M can be obtained as

$$M = bL(\text{CT}) \bmod n \tag{3}$$

where $L(\text{CT}) = \frac{\text{CT}^\lambda \bmod n^2 - 1}{n}$.

C. CP-ABE-Based Access Control Scheme

Attribute-based cryptography is a widely used encryption technique that supports fine-grained access [36]. It is divided into encryption key-policy attribute-based encryption (KP-ABE) [36] based on key policy attributes and encryption CP-ABE [25] based on ciphertext policy attributes. KP-ABE is an access policy embedded in the user key, and a specific access policy is assigned only when the Visitor needs to access the data. If the Visitor accesses multiple different data, he needs to have various keys that satisfy the match. Therefore, KP-ABE is suitable for static scenarios and is not applicable in adaptive protection and sharing scenarios. On the other hand, CP-ABE embeds the access policy into the ciphertext so that the Owner can formulate data access policies according to the user's identity attributes to achieve fine-grained access control to data. Simultaneously, the data access process is recorded in the Owner's tamper-evident private ledger to undertake data traceability.

The proposed access control method stores the data hash to consortium blockchain through CP-ABE, implements the Setup, KenGen, Encrypt, Upload, Download, and Decrypt operations of CP-ABE with the support of CA and utilizes DAPP and smart contract to complete the encryption and on-chain operations of the data hash. The proposed scheme requires the Owner to specify the user access control policy for encrypted data and generates ciphertext together with the access control policy and encrypted data hash. Only users who satisfy the access control policy can decrypt the ciphertext to obtain the hash of the encrypted data. At the same time, the CP-ABE method is implemented in CA that supervises certificate management in a consortium blockchain network and undertakes key initialization and distribution in the CP-ABE method. The following Table I describes the notation of the CP-ABE-based access control scheme.

Setup: The setup algorithm initializes the incoming system security parameter λ, generates the public parameters PK and a master key MK in CA

$$\text{Setup}\left(1^\lambda\right) \rightarrow (\text{MK}, \text{PK}). \tag{4}$$

KeyGen: Next, the user sends a registration request to CA, generating the user certificate CA_{Cert} and the user attribute private key SK_S corresponding to the private key SK. The UCR is a user certificate request that includes the public parameters PK of the user's certificate to be generated and the attribute S of the user. To guarantee the safety of the private key SK, CA, as a trusted third party, encrypts the private key SK according to the

Authorized licensed use limited to: SOUTHWEST JIAOTONG UNIVERSITY. Downloaded on March 01,2026 at 07:20:00 UTC from IEEE Xplore. Restrictions apply.

TABLE I
DESCRIPTION OF SYMBOLS

Symbols	Symbol Definition
λ	System security parameter
SK	The private key
SK_S	The user attribute private key
PK, MK	The public parameters and a master key
CA	Certificate Authority
Owner, Visitor	Owner and Visitor of the data
UCR	User certificate request
S	User's set of attributes
P	Data access control policies
$\mathrm{CA}_{\mathrm{Cart}}$	User certificates
M	The plaintext data
M_{hash}	The Owner's encrypted data hash
$\mathrm{CT}_{\mathrm{hash}}$	User data ciphertext hash
CT	The Owner's data ciphertext
$\mathrm{Tx(CT)}$	Transaction content on blockchain

public parameters PK corresponding to the user certificate in UCR and gets the user attribute private key SK_S corresponding to the private key SK. Therefore, the private key SK corresponding to the public parameters PK contained in the user certificate $\mathrm{CA}_{\mathrm{Cert}}$ is the only way to successfully receive the user attribute private key SK_S corresponding to the private key SK. After generating the SK_S, the CA returns $(\mathrm{SK}_S, \mathrm{CA}_{\mathrm{Cert}})$ to the user

$$\mathrm{KenGen(MK, UCR)} \rightarrow (\mathrm{SK}_S, \mathrm{CA}_{\mathrm{Cert}}). \tag{5}$$

Encrypt: The encrypt algorithm is mainly the interaction phase between the Owner and the blockchain network. The Owner specifies the access policy P and the encrypted data hash M_{hash} to be uploaded and encrypts the encrypted data hash M_{hash} with the access policy P and public parameters PK to generate the ciphertext CT

$$\mathrm{Encrypt}\ (\mathrm{PK}, M_{\mathrm{hash}}, P) \rightarrow \mathrm{CT}. \tag{6}$$

Upload: In this phase, the Owner initiates a transaction to the blockchain network and sends the ciphertext CT as the load $\mathrm{Tx(CT)}$ of the transaction to the on-chain for the blockchain network synchronization of the ciphertext data. The smart contract initiates the transaction and uses the ciphertext CT as the transaction content, namely $\mathrm{Tx(CT)}$. The transaction content $\mathrm{Tx(CT)}$ is uploaded to the blockchain network for synchronization through a secure channel, so the blockchain network endorses and verifies next the transaction $\mathrm{Tx(CT)}$. It should be noted that this method only verifies the format, signature, and status consistency of the transaction and does not verify the legitimacy of the specific data in the transaction, thus ensuring the privacy, confidentiality, and feasibility of ciphertext data CT on the blockchain

$$\mathrm{Upload(Tx(CT))} \rightarrow \text{ block } A. \tag{7}$$

Download: In the download algorithm, the Visitor on the blockchain wants to request access to the Owner's ciphertext CT through the blockchain network. Once block A containing the ciphertext transaction $\mathrm{Tx(CT)}$ is uploaded, the peer node in the whole channel will synchronize block A, so the Visitor requests the corresponding information containing the ciphertext of this transaction in the blockchain network through DAPP. The Visitor can obtain CT if the private key SK of the Visitor satisfies

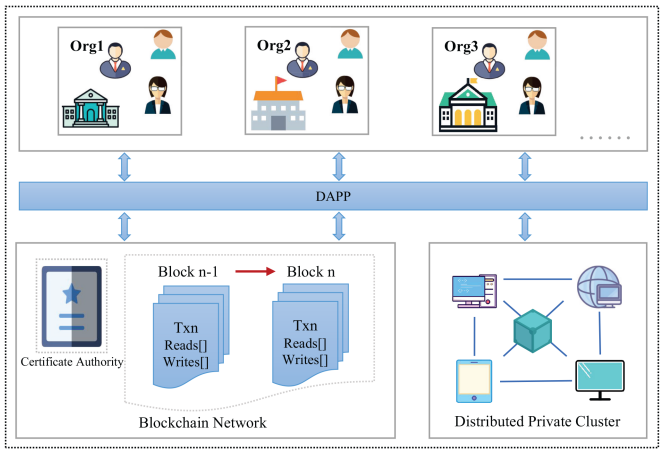


Fig. 2. Methodology model.

the requirement of P

$$\mathrm{Download(Tx(CT))} \rightarrow \mathrm{CT}. \tag{8}$$

Decrypt: Next, the Visitor decrypts the user attribute privacy key $\mathrm{SK_S}$ to obtain the privacy key SK in the CP-ABE access control method. Then, Visitor B decrypts the ciphertext of Owner A using the user attribute private key and the public parameters to get the encrypted data hash, thus realizing the access control of the data

$$\mathrm{Decrypt}\ (\mathrm{CT}, \mathrm{SK}_S, \mathrm{PK}) \rightarrow M_{\mathrm{hash}}. \tag{9}$$

We will use the above technologies to build our PDPChain solution model, where blockchain technology ensures data traceability and transaction transparency, and Paillier and CP-ABE cryptographic primitives ensure data privacy and security, enabling fine-grained access control centered on the data owner.

IV. CONSTRUCTION

In this section, we present the complete approach design that includes the PDPChain system architecture, PDPChain system components, the entire lifecycle of trusted storage, and the secure sharing process of data.

A. PDPChain System Architecture

To realize personal data security storage and reliable sharing. As shown in Fig. 2, the architecture of the approach proposed in PDPChain is mainly composed of five entities: 1) blockchain network, 2) distributed private cluster, 3) DAPP, 4) Owner, and 5) Visitor in collaboration.

1) Blockchain Network: Blockchain network ensures the trusted storage of private data through consortium blockchain and combined with the designed CP-ABE-based access control method to ensure fine-grained access control and secure and reliable data transmission on blockchain.

2) Distributed Private Cluster: PDPChain solves the problem of data storage size and speed on the blockchain by building a distributed private cluster and realizes the efficient and secure storage of data encrypted by the Paillier homomorphic encryption mechanism off-chain. Distributed private cluster stores the

Authorized licensed use limited to: SOUTHWEST JIAOTONG UNIVERSITY. Downloaded on March 01,2026 at 07:20:00 UTC from IEEE Xplore. Restrictions apply.

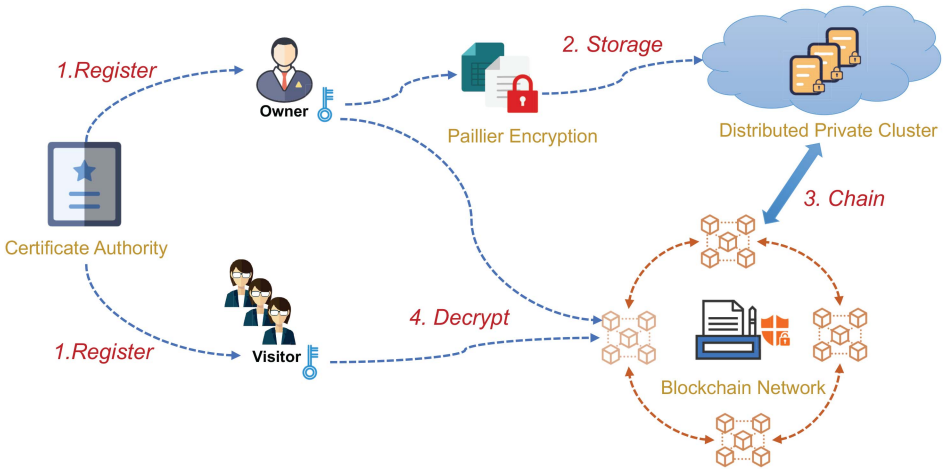


Fig. 3. PDPChain system components.

original information of Paillier encrypted data, then calculates and generates the hash corresponding to the encrypted data, and then uploads the data hash value to the consortium blockchain to achieve reliable storage and traceable access control.

3) Distributed Applications: DAPP runs on a blockchain network and can be a decentralized application protecting user privacy data, such as personal data management systems, file systems, and archive systems. To provide users with a good access window to consortium blockchain, consortium blockchain's underlying technology and applications are visualized through DAPP, making operations more transparent and surface. The system service provided smart contract and distributed private cluster interface calls for DAPP in the RESTful interface.

4) Owner and Visitor: In the system, Owner and Visitor can be node roles from the same or different organizations in the blockchain network. We consider the Owner of the data as the administrator of the data, who can upload, download, and share the data. The Owner of the data can upload the data to the distributed private cluster after encryption, encrypt the data with the data access control policy set in the blockchain network, and have the control and access the right of the data. The Visitor can initiate access requests to the Owner, and if their own attribute access set matches the access control policy of the data, they can access the data with the user private key.

B. System Components

As shown in Fig. 3, PDPChain consists of five phases: 1) Initialize, 2) Register, 3) Storage, 4) Chain, and 5) Decrypt.

Initialize: $\text{Initialize}(1^{\lambda}) \rightarrow (\text{PK}, \text{MK})$. The Owner and the Visitor input a security parameter λ, and the certificate authority CA in blockchain network returns the public parameters PK and a master key MK.

Register: $\text{Register}(\text{PK}, \text{S}, \text{MK}) \rightarrow (\text{SK}_S, \text{CA}_{\text{Cert}})$. With the PK and MK obtained in the Initialize phase, and the set of participant attributes S, the consortium blockchain-based trusted third-party CA generates the participant's certificate CA_{Cert}, and the user attribute private key SK_S in the CP-ABE access control method.

Storage: $\text{Storage}(M_{\text{hash}}) \rightarrow (\text{CT}_{\text{hash}})$. Based on the Paillier homomorphic encryption mechanism, the original data are encrypted and the ciphertext CT is stored in the distributed private cluster; then, the hash of the ciphertext CT is calculated by distributed private cluster. Finally, the ciphertext hash CT_{hash} is returned to the data owner through a secure channel.

Chain: $\text{Chain}(\text{CT}_{\text{hash}}) \rightarrow \text{block}A$. The ciphertext hash CT_{hash} encrypted based on the CP-ABE attribute is stored in the smart contract of consortium blockchain. Based on the RAFT consensus mechanism, by consensus the transaction to blockchain network, Endorser Peer verifies the signature, executes the transaction, and packages the transaction to Orderer after completion. Orderer sorts the transaction, generates the block, and broadcasts the block to all nodes, so finally, the Committer Peer synchronizes the block and saves the block to the ledger, completing the entire *Chain* process.

Decrypt: $\text{Decrypt}(\text{CT}_{\text{hash}}, \text{PK}, \text{SK}) \rightarrow M_{\text{hash}}$. Based on the CP-ABE access control scheme, if the attribute set S of the Visitor satisfies the access control policy P that requires access to the data file CT_{hash}, then M_{hash} can be obtained by its own PK and SK. The next step is to get the result CT of Paillier encryption by M_{hash} in distributed private cluster calculation.

Ultimately, decentralized storage for data, data ownership by data owners, fine-grained access control, and data availability invisibility can be achieved, truly enabling data transparency and auditability.

C. Full Life Cycle Process for Trusted Storage and Secure Sharing of Personal Data

To realize trusted storage and secure sharing of personal data on the blockchain, the proposed scheme is divided into the following six parts.

1) Paillier encrypts the original data.
2) Encrypted data are uploaded to distributed private cluster.
3) Data hash is trusted on blockchain.
4) On-chain personal data access control.
5) Distributed private cluster obtains encrypted data according to the hash.

Authorized licensed use limited to: SOUTHWEST JIAOTONG UNIVERSITY. Downloaded on March 01,2026 at 07:20:00 UTC from IEEE Xplore. Restrictions apply.

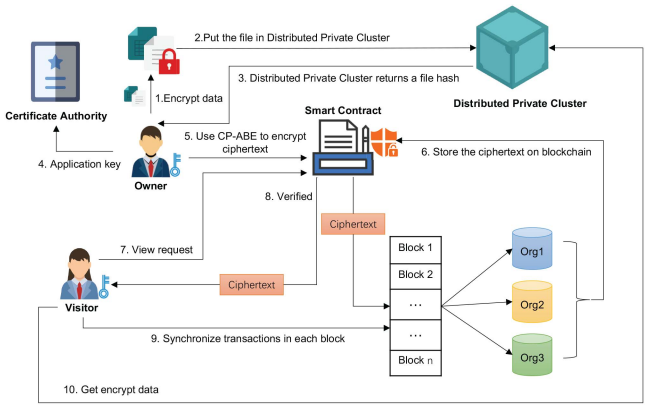


Fig. 4. Full lifecycle storage and sharing process of data.

Algorithm 1: paillierEncrypt(file).

// Encrypt the plaintext data using Paillier algorithm
system executes:
// Generate Paillier key
$n \leftarrow$ generate a public key;
$b \leftarrow$ generate a private key;
// read file content
$M \leftarrow$readFile$(file)$;
// encrypt
CT $\leftarrow$paillierEncrypt(n, M);
return n, CT

6) Storage of computation results.

Fig. 4 shows the whole lifecycle storage and sharing of data, and the implementation process is described in detail. As per the description, the data are encrypted by Paillier, then stored and shared as electronic files.

1) Paillier encrypts the original data: As in Fig. 4, the Owner in the organization initiates a Step 1 request to the encryption service via DAPP. First, the *PGenKen* algorithm randomly selects the public key $n = pq$ and calculates the secret key $b = \lambda^{-1} \mathrm{mod} n$. Then, the *PEncrypt* algorithm selects a random $r^n \mathrm{mod} n^2$ to encrypt the plaintext M. It outputs the ciphertext CT, and saves the secret key b by utilizing the encryption result as the file content. The algorithm for encrypting the plaintext data M through the Paillier homomorphic encryption mechanism is presented in Algorithm 1.

2) Encrypted data are uploaded to distributed private cluster: Personal data contain private information related to users in the Internet and secure storage is a very important part of PDPChain. Our PDPChain uses Merkle DAG of distributed private cluster and blockchain hash storage structure for secure storage. The distributed private cluster uses distributed hash tables (DHT) based on S/kademlia and Coral DSHT for content addressing, which enables content addressable, decentralized storage, and tamper-proof storage of personal data.

Following the same Fig. 4, the Owner of the organization initiates a Step 2 request to the distributed private cluster via DAPP. Upload their file data to the distributed private cluster via SDK. Distributed private clusters generate the Merkle DAG structure of the file data, and an arc policy caches the stored block. Then the legitimacy of the CID is checked. Finally, the storage path is stitched, and Put is invoked to keep the complete block. After distributed private cluster completes the hash calculation, the hash of the encrypted file is returned to the Owner via distributed private cluster SDK via Step 3.

3) Data hash is trusted on blockchain: Personal data are encrypted and stored in a distributed private cluster, and only the *Hash Index* generated by the distributed private cluster based on the ciphertext of the personal data is recorded in the blockchain. Blockchain uses the cryptographic hash function to achieve tamper-proof. Each organization has fully encrypted personal data, and any changes are recorded in the blockchain, thus enabling reliable storage of personal data.

The Owner sends a registration request to CA through Step 4. The CA runs the Setup(1^{λ}) algorithm to output the PK and the MK, which are stored in the Owner's entity. Next, the Owner sets the access policy to determine who is allowed to access the data. Then, the Owner uses the CP-ABE access control method through Step 5, and the Owner uses Encrypt $(\mathrm{PK}, M_{\mathrm{hash}}, P)$ algorithm in the CP-ABE access control method to encrypt the uploaded data hash and description to obtain the ciphertext CT.

Next, the smart contract runs the Upload$(\mathrm{Tx(CT)})$ algorithm and initiates a transaction with the ciphertext CT as the transaction content Tx(CT). The smart contract saves the ciphertext CT to the block, and the blockchain network executes Step 6 to perform consensus. After receiving the proposal, the Endorser peer verifies the signature and determines whether the Owner is entitled to operate. At the same time, it executes the smart contract according to the endorsement policy, sending the result and its respective CA certificate signature back to DAPP. After DAPP receives the information returned by the Endorser peer, it determines whether the proposal result is consistent and whether it is executed regarding the specified endorsement policy. If the Endorser peer verification fails, the transaction is aborted; otherwise, the DAPP client packages the data into a transaction and signs it, sending it next to the Orderer. The Orderer performs RAFT consensus sorting on the received transactions and generates a new block to be sent to the Committer peer. After obtaining the block, the Committer peer first checks whether the input and output dependent on each transaction in the block match the current state of the blockchain. If it does, the block is synchronized, and the ledger is formed.

The algorithm for uploading file data to distributed private cluster and file hash trusted storage blockchain is shown in Algorithm 2. The Owner uses the certificate $\mathrm{CA}_{\mathrm{Cert}}$ issued by CA to join the consortium blockchain, so the Owner uploads file information fileName and other info through DAPP. Next, the Owner encrypts the file data with Paillier, uploads the encrypted result to the distributed private cluster, and obtains the returned hash. The Owner initiates a request to CA to get the public parameters PK and the master key MK. The ciphertext CT is obtained by encrypting the file data using the CP-ABE access control policy P. The blockchain network stores the ciphertext, file relationship records, and operation records in the smart contract EFileInfo, UserEFileRelation, and OperationRecord, respectively, and finally returns the public parameters PK and file ID.

Authorized licensed use limited to: SOUTHWEST JIAOTONG UNIVERSITY. Downloaded on March 01,2026 at 07:20:00 UTC from IEEE Xplore. Restrictions apply.

Algorithm 2: uploadFile($\mathrm{CA_{Cert}}$,fileName,otherInfo,fileCT,P)

// upload files
system executes:
$id \leftarrow$ generate a global id;
// Save files to distributed private cluster
$hash \leftarrow$saveFileTodpc($fileCT$);
// Get key based on attribute request
MSP, PK, MK $\leftarrow$requestKeyFromCa(P);
$file \leftarrow File(id, fileName, hash, otherInfo)$
// hash encryption result
$\mathrm{CT_{hash}} \leftarrow$cpabeEncrypt(MK, PK, $file$);
contract executes:
// Get user information
$user \leftarrow$getFromCA($\mathrm{CA_{Cert}}$);
// File information data on blockchain
invokeSave(EFileInfo, id, $\mathrm{CT_{hash}}$);
// File Owner information data on blockchain
invokeSave(UserEFileRelation, $user$, id);
// File Owner information data on blockchain
invokeSave(OperaRecord, $user$, id, 2);
system executes:
return PK, id

Algorithm 3: downloadFile(fileId, CAcert).

// Download file
system executes:
SK_S, PK $\leftarrow$getFromCA($\mathrm{CA_{Cert}}$, $fileId$);
SK $\leftarrow$caCertDecode(SK_S);
contract executes:
// File operation record data on blockchain
save(OperaRecordCC, $user$, id, 1);
// Query file information
$\mathrm{CT_{fileInfo}}$, $error \leftarrow$query(EFileInfo, $fileId$);
if $error \neq null$:
return null, $error$("File does not exist!");
// CP-ABE authentication
$error \leftarrow$auth(PK, SK_S, $\mathrm{CT_{fileInfo}}$);
if $error \neq null$:
return $error$("Limits of authority!");
else:
return $\mathrm{CT_{fileInfo}}$;
system executes:
// CP-ABE decryption
$fileInfo, error \leftarrow$decrypt(PK, SK_S, $\mathrm{CT_{fileInfo}}$);
if $error \neq null$:
return $error$("Limits of authority!");
else:
$hash \leftarrow$fileInfo.hash;
$fileContent \leftarrow$downloadFileFromdpc($hash$);
return $fileInfo, fileContent$;

4) On-chain personal data access control: To enhance the interoperability and circulation of personal data in the network, the reliable sharing of personal data is also very necessary. The Visitor in the blockchain network initiates a Step 7 view request transaction to the smart contract and submits its own attribute set S. The Proposal sends to the Endorse peer the contract identifier to be invoked for this transaction: the contract method, the Visitor's transaction information file ID, the Visitor's attribute set S, and signature.

Once the Endorse peer receives the Proposal, it verifies the Visitor's signature and then checks via Step 8 that the attribute set S of the Visitor matches the access control list P of the requested file data. If it does not fit, the transaction is abandoned, and the processing is terminated. On the other hand, the smart contract is executed according to the Endorse policy simulation if it matches. Send the ciphertext CT to the Visitor by running the Download(Tx(CT)) algorithm, and CA certificates of the Owner and the Visitor are signed and then issued to DAPP. The Visitor runs the Decrypt(CT, SK_S, PK) algorithm and decrypts the ciphertext CT using the PK and the user attribute private key SK_S to obtain the file data hash M_{hash}.

The Visitor requests through Step 9, Orderer peer performs RAFT consensus sorting on the transaction, packages the contents of the transaction, and generates a new block to seed to Committer peer. After receiving the block again, Committer peer checks the transaction, checks the transaction dependent input and output after completion, the block is appended to blockchain and the world state is changed, and the transaction records are synchronized to each block of the blockchain network.

5) Distributed private cluster obtains encrypted data according to the hash: The Visitor in the blockchain network initiates a Step 10 request to the distributed private cluster. All nodes connected to it will form a private network for the current node. Thus, whenever the Visitor initiates a Get request to distributed private cluster, distributed private cluster first checks whether the requested data exist in the local block store. If it does, it returns it directly from the local block store; otherwise, the peer node initiates a block hash list to the private network, finds the node where each block hash is located through DHT and routing layer algorithms, and returns the encrypted data. The local node caches a copy of the block file to the local block store while receiving it and then updates the DHT, so finally, it assembles the block file and returns the encrypted data to the Visitor.

The file data access algorithm is depicted in Algorithm 3. The Visitor uses the certificate $\mathrm{CA_{Cert}}$ issued by CA to join consortium blockchain, the Visitor sends the file ID requested to be viewed through DAPP and obtains the user attribute private key encrypted with CA certificate and the public parameters PK from CA. Next, the Visitor decrypts SK_S using the CA certificate to retrieve the private key SK. The Visitor obtains the file information ciphertext EncryptText according to the file ID and verifies the Visitor's identity. If the verification passes, the Visitor will use the private key SK_S to decrypt the ciphertext and obtain the file data Hash and data description fileInfo. On the other hand, if the verification does not pass, the permission prompt message is returned after passing the request to distributed private cluster to retrieve the file content fileContent.

Authorized licensed use limited to: SOUTHWEST JIAOTONG UNIVERSITY. Downloaded on March 01,2026 at 07:20:00 UTC from IEEE Xplore. Restrictions apply.

6) Storage of computation results: The *PDecrypt* algorithm input the ciphertext CT. It outputs the plaintext M. As shown in Fig. 4, after the third-party platform extracts the data to the end of the calculation in Step 8, the calculation result is saved to the system. The Owner can decrypt the result and obtain the calculation results.

V. Experimental Results

In this section, we discuss the security and privacy of PDPChain and give a performance test analysis to demonstrate its advantages of high availability, precise role functionality, fine-grained access control, and high performance in the storage and sharing of personal data.

A. Security and Privacy Analysis

1) Paillier homomorphic encryption mechanism*:* If the following equation holds, it can be inferred that the Paillier homomorphic encryption mechanism is correct

$$\begin{aligned} &E(M) \times E(M')^{-1} \\ &= ((1+mn)\, r^n) \times \left((1+m'n)\, r'^n\right)^{-1} \mathrm{mod} n^2 \\ &= \left(1 + \frac{n\,(m-m')}{1+nm'}\right) \times \left(\frac{r}{r'}\right)^n \mathrm{mod} n^2. \end{aligned}$$

If $M = M'$, it can be derived that $D(E(M) \times E(M')^{-1}) = D(1) = 0$. From this, the ciphertext $E(M)$ after encryption by Paillier can be calculated.

Data Security: The Visitor cannot infer the plaintext data $M_1, M_2, \ldots, M_n$ from $E(M_1), E(M_2), \ldots, E(M_n)$. The Owner stores the plaintext data M encrypted in the DHT of the distributed private cluster, and the last data accessed by the Visitor is the data $E(M_1), E(M_2), \ldots, E(M_n)$ encrypted by Paillier. If the Visitor wants to access the original data or additional information, the plaintext data M are obtained by $D(E(M) \times E(M')^{-1}/E(M')^{-1}))$. However, since the Owner stores the computation result of the plaintext data M unordered in the DHT, the likelihood of the observed data is equal. If there are n data, the probability of correct guess is $1/n$, so the plaintext data M is secure, and the Visitor cannot infer the Owner's private data.

2) CP-ABE-based access control method:

a) Data Privacy Protection: The original data are encrypted by Paillier and uploaded to distributed private cluster, so the hash of the generated data is calculated. The Owner only encrypts the hash by CP-ABE encryption technology to upload blockchain, and thus, only the ciphertext CT of the encrypted hash is eyeable to the users in the blockchain network. Users who comply with the access control policy P can decrypt and access the ciphertext CT with their keys, and the Visitor who does not adhere to the access control policy P cannot decrypt and access the ciphertext CT. Thus, the privacy of private data is warranted.

b) User-centric: In the proposed scheme, the Owner can generate public parameters PK and the system master key MK, and generate and distribute the user attribute private key SK_S for the Visitor based on its attribute set S. The Owner controls this process; thus, it has complete control over his shared data.

c) Fine-grained access control: In the proposed scheme, fine-grained access control of shared data is implemented through the CP-ABE access control scheme. The Owner can autonomously set the access control policy P and embed the Visitor's attribute set S into the ciphertext CT.

d) Secure key distribution: The private key SK generated in the key distribution phase is trusted, because the public parameters PK and the master key MK are generated in the fully trusted CA, and then, the user attribute private key SK_S is generated by the trusted master key MK and the user attributes set S. Meanwhile, it is not the private key SK that is transmitted and circulated in the blockchain network, but the user attribute private key SK_S generated by encrypting private key with the public parameters PK that is transmitted and distributed. The Visitor who conforms to the access control policy P can only decrypt the hash and file description in SK_S and cannot acquire the private key SK of the Owner.

e) Safe storage and sharing of data: The security of storing and sharing is an important feature of PDPChain. In this scheme, the Owner can have complete control over their own data. The process from data storing to data sharing is all secure. During the data storing, the Owner encrypts the original data using Paillier homomorphic encryption technology in a distributed private cluster. During the data release, the indexes of data are reserved in the consortium blockchain. Moreover, blockchain is a distributed database without SPOF and each node has a copy of transaction records. Besides, the digital signature provides authentication, integrity, and nonrepudiation for each transaction. During the data sharing, the data access permissions are preset in the smart contract. Only authorized users and institutions can use the data. The executed access records are stored in the blockchain to trace the behaviors of data.

3) Security and privacy comparison: Since the proposed PDPChain is implemented in a decentralized blockchain, compared with the traditional cloud storage schemes [9], [10], the PDPChain scheme can avoid SPOF, and, at the same time, solve the problems of centralized storage, third-party data leakage. Users do not have issues like data ownership. Both the consortium blockchain and the distributed private cluster used in PDPChain have distributed technologies. Second, blockchain is based on RAFT consensus protocol, so thus, even if some nodes fail, it cannot affect the availability of the overall solution.

We compare the proposed PDPChain scheme with schemes [4], [8], [19], [21] in terms of data security, data privacy protection, user-centric, fine-grained access control, and secure key distribution, respectively, and the results of the security analysis are shown in Table II.

The massive data existing in the Internet are suffering from third-party cloud computing and personal privacy leakage. In order to provide effective privacy protection of personal data, a blockchain technology based on distributed ledger is urgently needed, but the plaintext is stored on the blockchain. Data and all trusted nodes can see the data; it is urgent to use encryption technology to protect data privacy and perform

Authorized licensed use limited to: SOUTHWEST JIAOTONG UNIVERSITY. Downloaded on March 01,2026 at 07:20:00 UTC from IEEE Xplore. Restrictions apply.

TABLE II
SAFETY COMPARISON

Comparative aspects	Ours	Pro. [4]	Pro. [8]	Pro. [19]	Pro. [21]
Data security	✓	×	×	×	×
Data privacy protection	✓	✓	✓	✓	×
User-centric	✓	✓	×	✓	✓
Fine-grained access control	✓	✓	✓	✓	✓
Secure key distribution	✓	✓	✓	×	✓

Note: "✓" indicates that the literature scheme supports this feature.
"×" indicates that the literature scheme does not support this feature.

fine-grained access control to data based on users. As shown in Table II, compared to other schemes in different aspects, such as data security, data privacy protection, user-centric, fine-grained access control of data, and secure key distribution, the proposed scheme has better security performance and is more suitable for trusted storage and secure sharing of personal privacy data in the consortium blockchain.

B. Performance Analysis

In the performance test environment setup, our experimental environment configuration used for PDPChain's blockchain network is Intel Core i5-5257U CPU, 8 GB memory, running operating system Ubuntu (18.04 version). According to the current popular technology and the technical architecture that fits with the solution, Hyperledger Fabric is used to build the consortium blockchain and IPFS is used for the distributed private cluster. The blockchain network is deployed using Hyperledger Fabric 1.4 and consists of two organizations, each composed of two peer nodes and two users. The IPFS cluster consists of a privatized cluster of three servers with the same configuration on an Ubuntu system (16.04 version), with an Intel Core i5-9400F CPU and 3 GB memory. The data transfer bandwidth between all servers is 100 Mbps. In addition, the RESTful service uses the Gin framework and the Fabric-SDK-GO platform, using the Go to write the relevant code.

In the performance test, the original data are encrypted by Paillier homomorphic encryption mechanism and then saved through file form. We select the following four key steps for performance testing:

1) Paillier encryption and decryption time consuming;
2) data upload to IPFS to get hash;
3) file hash encryption on blockchain;
4) access control for data on blockchain.

The time consumption of each step is tested and measured.

1) Paillier encryption and decryption time consuming: The data are encrypted before uploading it to IPFS and testing the speed of encryption and decryption of the original Paillier algorithm [26] in this step compared to the improved algorithm used in this article. Fig. 6 illustrates that the overall time reduction in encryption and decryption is 25%, and Fig. 5 displays that the improved algorithm reduces the encryption time by 48%. From the results in Figs. 5 and 6, through the improved Paillier algorithm, only two multiplication operations are required for encryption, which further improves the efficiency and efficiency of the system compared to the original Paillier algorithm.

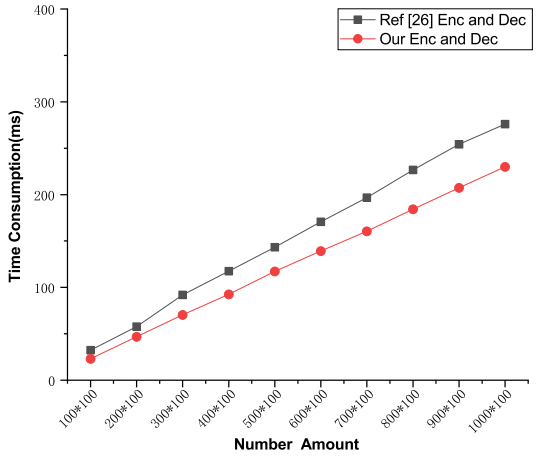


Fig. 5. Paillier encryption time overhead.

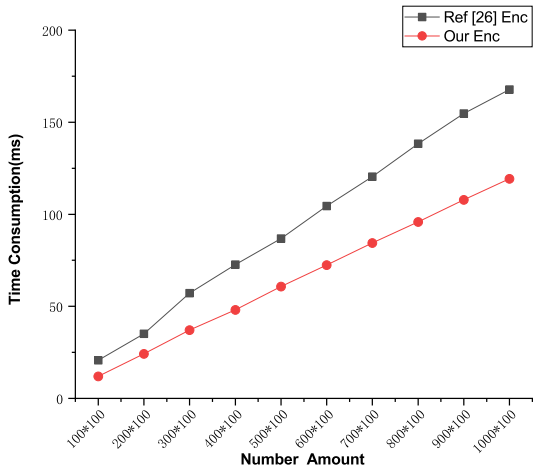


Fig. 6. Paillier encryption and decryption time overhead.

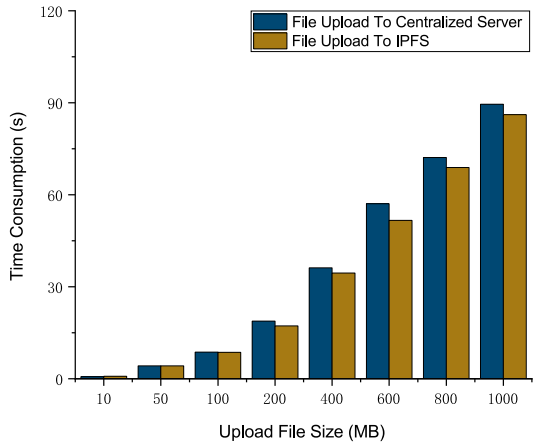


Fig. 7. File upload time.

2) Data upload to IPFS to get hash: The traditional data storage is that users upload data to the centralized server, and the proposed PDPChain scheme is that nodes upload data in the form of files to IPFS to generate the hash. Further experimental tests are conducted to compare the experiments of uploading files of different sizes to IPFS and centralized servers. The experimental results are shown in Fig. 7. As can be seen from Fig. 7, the time

Authorized licensed use limited to: SOUTHWEST JIAOTONG UNIVERSITY. Downloaded on March 01,2026 at 07:20:00 UTC from IEEE Xplore. Restrictions apply.

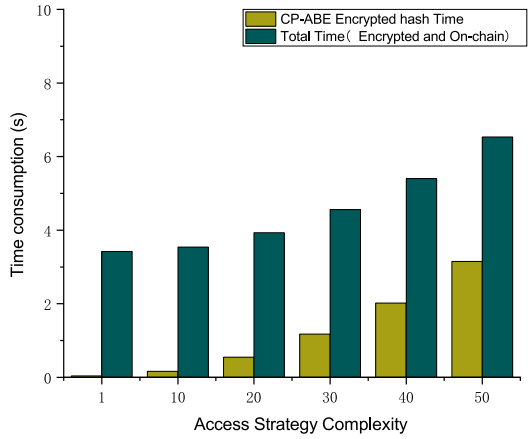


Fig. 8. File encryption upload time.

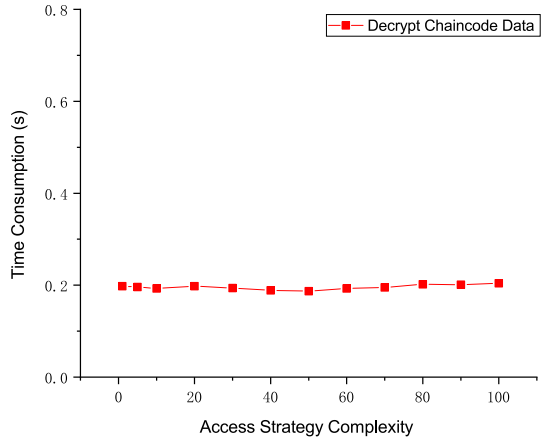


Fig. 9. Hash decryption time is independent of the access control policy.

cost of both storage methods increases linearly, and even for a file size of 1000 MB, the time to upload a local file to the IPFS private cluster is about 86 s, and the difference in time cost with the centralized server is about 3 s. Although the time taken to upload the same size file to the centralized server is slightly higher than the time taken to upload to IPFS in the same bandwidth environment, the overall time cost of uploading is still within an acceptable range, showing that the introduction of IPFS does not cause additional time overhead to the system.

3) File hash encryption upload blockchain: The node encrypts the data hash and data, such as the access policy, then uploads it. The node first requests a key from Fabric-CA and then encapsulates the information such as hash, data access policy, and data description in the ciphertext. Then, the ciphertext is encrypted and uploaded to the blockchain network.

As the access policy complexity increases, the encryption process time consumption also increases, and therefore, the relationship between policy complexity and encryption time consumption is tested in the experiment. In the proposed method, the access control policy complexity is set to three latitudes to ensure the security of data and the efficiency of access control by combining the characteristics of Fabric. As shown in Fig. 8 is the whole file encryption upload time, which is the sum of certificate acquisition from Fabric-CA, encryption time consumption, and upload time consumption. Observing the data in the figure, the time cost of the encryption scheme based on CP-ABE is very small in the entire chain stage, and the increase of time is with the increase of the complexity of the access policy.

4) Access control for data on blockchain: The entire data access control process includes obtaining the ciphertext CT from the blockchain network, decrypting the ciphertext CT to get the M_{hash}, and retrieving the Paillier encrypted data from IPFS, according to the hash.

Since the content stored on blockchain is of the same size, the file hash length is fixed, i.e., the length of the data on the blockchain is fixed. From this point, as the time taken to obtain the ciphertext from the blockchain network is fixed, the time taken is not tested. As depicted in Fig. 9, the decryption speed is independent of the policy complexity. As policy complexity of the decryption process, the decryption time consumption slightly changed with an increase in the complexity of the access policy in testing. The time cost is stable at about 0.2 s, and the overall time consumption is minimal.

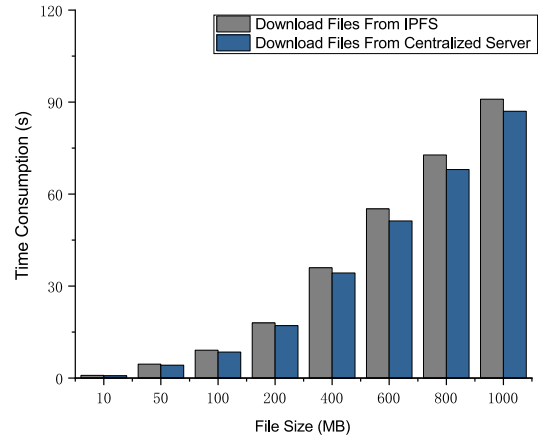


Fig. 10. Downloading file data from IPFS takes about the same amount of time as downloading file data from a centralized server.

Fig. 10 tests the time cost of downloading file data from IPFS using IPFS-Go-SDK based on hash versus downloading file data from a centralized server. It can be observed that downloading file data from IPFS and downloading it from a centralized server takes approximately the same amount of time.

To better compare the overall time consumption of the PDPChain model in terms of the time consumption of each step, the overall time overhead of the test scenario is combined with the actual situation. The access policy complexity applied in this article is 3, the Paillier homomorphic encrypted data volume of 100 000, and an upload and download file size of 600 MB for the overall time consumption test. The experimental results are shown in Fig. 11. It is seen that the highest percentage of time consumption in the complete scheme is the file transfer process, also improvements in the file transfer speed. However, it can be seen that the data uploading and encryption/decryption time overhead in the improved encryption technology is minimal, the same as the overhead of the complete scheme.

Overall, the results obtained point out that the proposed scheme has the same performance as the traditional centralized

Authorized licensed use limited to: SOUTHWEST JIAOTONG UNIVERSITY. Downloaded on March 01,2026 at 07:20:00 UTC from IEEE Xplore. Restrictions apply.

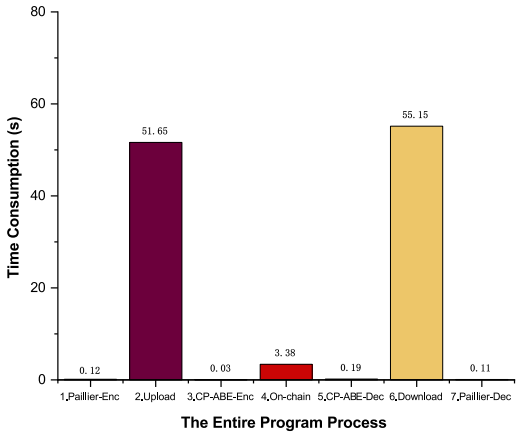


Fig. 11. Overall time overhead of the scheme.

storage scheme for data, and there is no performance degradation due to the introduction of CP-ABE and Paillier's homomorphic encryption mechanism. In contrast, this proposed scheme has an optimized speed advantage when uploading large data files.

VI. Conclusions and Future Directions

In the AI-driven era, to solve problems such as privacy leakage and realize granular access control in cloud computing, we proposed PDPChain, a consortium blockchain-based approach to personal data privacy protection. In this scheme, the encrypted data were stored through the distributed private cluster, and a hash was calculated and transmitted on the blockchain to improve the storage rate on blockchain. In addition, the data circulating in the scheme were encrypted by the improved Paillier homomorphic encryption mechanism to achieve secure audit and available invisibility of the data. Experimental analysis indicated that the proposed scheme has the advantages of high availability, high security, clear role function, fine-grained access control, and high performance, achieving the improvement and solution of the problems of trusted storage and secure sharing of personal data with lower usage overhead and higher decoupling of the system.

In future directions for this investigation, we continue to delve into issues related to the privacy computation of personal data, discuss and utilize a better algorithm for federated learning and homomorphic encryption to optimize the proposed scheme further and more comprehensively address the privacy and security of data in blockchain.

References

[1] D. Yoon, S. Moon, K. Park, and S. Noh, "Blockchain-based personal data trading system using decentralized identifiers and verifiable credentials," in *Proc. Int. Conf. Inf. Commun. Technol. Convergence*, 2021, pp. 150–154.

[2] S. R. K. Somayaji, M. Alazab, M. K. Manoj, A. Bucchiarone, C. L. Chowdhary, and T. R. Gadekallu, "A framework for prediction and storage of battery life in IoT devices using DNN and blockchain," in *Proc. IEEE Globecom Workshops*, 2020, pp. 1–6.

[3] W. Liang, S. Xie, D. Zhang, X. Li, and K.-C. Li, "A mutual security authentication method for RFID-PUF circuit based on deep learning," *ACM Trans. Internet Technol.*, vol. 22, no. 2, 2021, Art. no. 34.

[4] H. Gao, Z. Ma, S. Luo, Y. Xu, and Z. Wu, "BSSPD: A blockchain-based security sharing scheme for personal data with fine-grained access control," *Wireless Commun. Mobile Comput.*, vol. 2021, 2021, Art. no. 6658920.

[5] W. Liang et al., "Deep neural network security collaborative filtering scheme for service recommendation in intelligent cyber-physical systems," *IEEE Internet Things J.*, to be published, doi: 10.1109/JIOT.2021.3086845.

[6] W. Liang, Y. Li, J. Xu, Z. Qin, and K. Li, "QoS prediction and adversarial attack protection for distributed services under DLaaS," *IEEE Trans. Comput.*, to be published, doi: 10.1109/TC.2021.3077738.

[7] W. Liang, S. Xie, J. Cai, C. Wang, Y. Hong, and X. Kui, "Novel private data access control scheme suitable for mobile edge computing," *China Commun.*, vol. 18, no. 11, pp. 92–103, 2021.

[8] S. Wang, Y. Zhang, and Y. Zhang, "A blockchain-based framework for data sharing with fine-grained access control in decentralized storage systems," *IEEE Access*, vol. 6, pp. 38437–38450, 2018.

[9] J. R. Gudeme, S. K. Pasupuleti, and R. Kandukuri, "Attribute-based public integrity auditing for shared data with efficient user revocation in cloud storage," *J. Ambient Intell. Humanized Comput.*, vol. 12, no. 2, pp. 2019–2032, 2021.

[10] G. Nagarajan and K. S. Kumar, "Security threats and challenges in public cloud storage," in *Proc. Int. Conf. Adv. Comput. Innov. Technol. Eng.*, 2021, pp. 97–100.

[11] W. Liang, L. Xiao, K. Zhang, M. Tang, D. He, and K.-C. Li, "Data fusion approach for collaborative anomaly intrusion detection in blockchain-based systems," *IEEE Internet Things J.*, to be published, doi: 10.1109/JIOT.2021.3053842.

[12] G. Zyskind, O. Nathan, and A. S. Pentland, "Decentralizing privacy: Using blockchain to protect personal data," in *Proc. IEEE Secur. Privacy Workshops*, 2015, pp. 180–184.

[13] C. Zhang, Y. Xu, Y. Hu, J. Wu, J. Ren, and Y. Zhang, "A blockchain-based multi-cloud storage data auditing scheme to locate faults," *IEEE Trans. Cloud Comput.*, to be published, doi: 10.1109/TCC.2021.3057771.

[14] E. Androulaki et al., "Hyperledger fabric: A distributed operating system for permissioned blockchains," in *Proc. 13th EuroSys Conf.*, 2018, pp. 1–15.

[15] G. Wood et al., "Ethereum: A secure decentralised generalised transaction ledger," *Ethereum Project Yellow Paper*, vol. 151, no. 2014, pp. 1–32, 2014.

[16] V. Buterin, "A next-generation smart contract and decentralized application platform[J]." White Paper, vol. 3, no. 37, pp. 1–2, 2014

[17] X. Xu et al., "Latency performance modeling and analysis for hyperledger fabric blockchain network[J]," *Inf. Process. Manage.*, vol. 58, no. 1, 2021, Art. no. 102436.

[18] A. Azaria, A. Ekblaw, T. Vieira, and A. Lippman, "MedRec: Using blockchain for medical data access and permission management," in *Proc. 2nd Int. Conf. Open Big Data*, 2016, pp. 25–30.

[19] J. Grabis, V. Stankovski, and R. Zaričš, "Blockchain enabled distributed storage and sharing of personal data assets," in *Proc. IEEE 36th Int. Conf. Data Eng. Workshops*, 2020, pp. 11–17.

[20] F. Li, K. Liu, L. Zhang, S. Huang, and Q. Wu, "EHRChain: A blockchain-based EHR system using attribute-based and homomorphic cryptosystem," *IEEE Trans. Serv. Comput.*, to be published, doi: 10.1109/TSC.2021.3078119.

[21] R. Cong, Y. Liu, K. Tago, R. Li, H. Asaeda, and Q. Jin, "Individual-initiated auditable access control for privacy-preserved IoT data sharing with blockchain," in *Proc. IEEE Int. Conf. Commun. Workshops*, 2021, pp. 1–6.

[22] J. Liu, X. Li, L. Ye, H. Zhang, X. Du, and M. Guizani, "BPDS: A blockchain based privacy-preserving data sharing for electronic medical records," in *Proc. IEEE Glob. Commun. Conf.*, 2018, pp. 1–6.

[23] L. Tan, K. Yu, N. Shi, C. Yang, W. Wei, and H. Lu, "Towards secure and privacy-preserving data sharing for COVID-19 medical records: A blockchain-empowered approach," *IEEE Trans. Netw. Sci. Eng.*, vol. 9, no. 1, pp. 271–281, Jan./Feb. 2022.

[24] L. Li, Z. Yue, and G. Wu, "Electronic medical record sharing system based on hyperledger fabric and interplanetary file system," in *Proc. 5th Int. Conf. Compute Data Anal.*, 2021, pp. 149–154.

[25] J. Bethencourt, A. Sahai, and B. Waters, "Ciphertext-policy attribute-based encryption," in *Proc. IEEE Symp. Secur. Privacy*, 2007, pp. 321–334.

[26] P. Paillier, "Public-key cryptosystems based on composite degree residuosity classes," in *Proc. Int. Conf. Theory Appl. Cryptographic Techn.*, 1999, pp. 223–238.

[27] S. Ma, Y. Deng, D. He, J. Zhang, and X. Xie, "An efficient NIZK scheme for privacy-preserving transactions over account-model blockchain," *IEEE Trans. Dependable Secure Comput.*, vol. 18, no. 2, pp. 641–651, Mar./Apr. 2021.

[28] B. Bünz, S. Agrawal, M. Zamani, and D. Boneh, "Zether: Towards privacy in a smart contract world," in *Proc. Int. Conf. Financial Cryptography Data Secur.*, 2020, pp. 423–443.

Authorized licensed use limited to: SOUTHWEST JIAOTONG UNIVERSITY. Downloaded on March 01,2026 at 07:20:00 UTC from IEEE Xplore. Restrictions apply.

[29] C. Piao, Y. Hao, J. Yan, and X. Jiang, "Privacy preserving in blockchain-based government data sharing: A service-on-chain (SOC) approach," *Inf. Process. Manage.*, vol. 58, no. 5, 2021, Art. no. 102651. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0306457321001400

[30] V. K. A. Sandor, Y. Lin, X. Li, F. Lin, and S. Zhang, "Efficient decentralized multi-authority attribute based encryption for mobile cloud data storage," *J. Netw. Comput. Appl.*, vol. 129, pp. 25–36, 2019. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S1084804519300037

[31] Y. Qiu, H. Zhang, Q. Cao, and J. Zhang, "Blockchain data access control scheme based on CP-ABE algorithm," *Chin. J. Netw. Inf. Secur.*, vol. 6, no. 3, pp. 88–98, 2020.

[32] M. Poongodi, M. Hamdi, V. Varadarajan, B. S. Rawal, and M. Maode, "Building an authentic and ethical keyword search by applying decentralised (blockchain) verification," in *Proc. IEEE Conf. Comput. Commun. Workshops*, 2020, pp. 746–753.

[33] Z. Guan, X. Lu, W. Yang, L. Wu, N. Wang, and Z. Zhang, "Achieving efficient and privacy-preserving energy trading based on blockchain and ABE in smart grid," *J. Parallel Distrib. Comput.*, vol. 147, pp. 34–45, 2021.

[34] D. Ongaro and J. Ousterhout, "In search of an understandable consensus algorithm," in *Proc. USENIX Annu. Tech. Conf.*, 2014, pp. 305–319. [Online]. Available: https://www.usenix.org/conference/atc14/technical-sessions/presentation/ongaro

[35] Z. Jiang, J. Liu, and Y. Wang, "Improvement on Paillier-pointcheval probabilistic public-key encryption scheme," *Comput. Eng.*, vol. 34, no. 3, pp. 38–39, 2008.

[36] D. Han, Y. Zhu, D. Li, W. Liang, A. Souri, and K.-C. Li, "A blockchain-based auditable access control system for private data in service-centric IoT environments," *IEEE Trans. Ind. Informat.*, vol. 18, no. 5, pp. 3530–3540, May 2022.



Wei Liang received the Ph.D. degree in computer science and technology from Hunan University, Changsha, China, in 2013.

He was a Postdoctoral Scholar with Lehigh University during 2014–2016. He is currently a Professor and the Dean with the School of Computer Science and Engineering, Hunan University of Science and Technology, Xiangtan, China. He has authored or coauthored more than 140 journal/conference papers such as the IEEE TRANSACTIONS ON INDUSTRIAL INFORMATICS, IEEE TRANSACTIONS ON EMERGING TOPICS IN COMPUTING, IEEE TRANSACTIONS ON COMPUTATIONAL BIOLOGY AND BIOINFORMATICS, and *IEEE Internet of Things Journal*. His research interests include blockchain security technology, networks security protection, embedded system and hardware IP protection, fog computing, and security management in wireless sensor networks.



Yang Yang received the B.Eng. degree in computer science and technology form Hunan Normal University, Changsha, China, in 2021. She is currently working toward the Graduate degree in electronic information with the College of Computer Science and Electronic Engineering, Hunan University, Changsha, China.

Her research interests include blockchain security technology, privacy protection, and data security and privacy.



Ce Yang received the Ph.D. degree in information and communication engineering from the University of Science and Technology of China, Hefei, China, in 2017.

He is currently a Ph.D. Lecturer with the School of Computer science and Engineering, Hunan University of Science and Technology, Xiangtan, China. His research interests include blockchain, privacy protection, and distributed computing.



Yonghua Hu received the Ph.D. degree in computer application technology from Hunan University, Changsha, China, in 2008.

He is currently a Professor and the Vice Dean with the School of Computer Science and Engineering, Hunan University of Science and Technology, Xiangtan, China. His research interests include compiler, parallel computing, and trusted computing.



Songyou Xie received the B.Sc. degree in optical information science and technology and the M.S. degree in computer science and technology from the Hunan University of Science and Technology, Xiangtan, China, in 2016 and 2019, respectively. He is currently working toward the Ph.D. degree with Hunan University, Changsha, China.

His research interests include blockchain and lightweight security authentication protocols.



Kuan-Ching Li (Senior Member, IEEE) received the Ph.D. degree in electrical engineering from the University of São Paulo, São Paulo, Brazil, in 2001.

Besides publication in high-quality conferences and journals, he is a coauthor or coeditor of more than 40 books published by Taylor & Francis, Springer, Elsevier, and McGraw-Hill. His research interests include parallel and distributed computing, big data, blockchain, and emerging technologies.

Dr. Li is a fellow of the IET and a Member of the AAAS.



Jiannong Cao (Fellow, IEEE) received the Ph.D. degree in computer science from Washington State University, Pullman, WA, USA, in 1990.

He is currently the Otto Poon Charitable Foundation Professor in data science and the Chair Professor of Distributed and Mobile Computing with the Department of Computing, The Hong Kong Polytechnic University (PolyU), Hong Kong. His research interests include distributed systems and blockchain, wireless sensing and networking, big data and machine learning, and mobile cloud and edge computing.

Prof. Cao is a member of Academia Europaea, a fellow of China Computer Federation, and an ACM Distinguished Member.

Authorized licensed use limited to: SOUTHWEST JIAOTONG UNIVERSITY. Downloaded on March 01,2026 at 07:20:00 UTC from IEEE Xplore. Restrictions apply.